News Sentiment and Directional Price Forecasting

A Comparative Study of Technology vs Financial Sectors

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Abstract

This paper investigates whether news sentiment derived from the New York Times can improve directional stock price prediction, and whether this effect differs between the technology sector (XLK) and the financial sector (XLF). Using five years of daily data (January 2021 – February 2026), we construct a sentiment analysis pipeline based on the Loughran-McDonald (2011) financial dictionary and train four custom text-classification architectures. Thirty predictive features — spanning technical indicators, sentiment derivatives, macro-economic variables, and trading volume — are evaluated using five models: XGBoost, LSTM, Logistic Regression, Random Forest, and ARIMA, all assessed under a rigorous rolling walk-forward backtest with 0.1% transaction costs.

We find that H1 (XLK exhibits higher sentiment-driven predictability than XLF) is validated: a daily-level paired t-test — pairing by trading day to control for shared macroeconomic conditions — (n=1,010, t=2.747, p=0.003) confirms XLK's directional accuracy significantly exceeds XLF's. Granger causality tests corroborate the mechanism — sentiment Granger-causes XLK returns at lags 1–4 days (6 features significant, p<0.05) but shows no causal relationship for XLF. H2 (sentiment features significantly improve accuracy) is rejected across all model–sector combinations. Paradoxically, XLF's model relies more heavily on sentiment features (41.7% vs 38.8%), yet cannot translate this reliance into predictive gains. We interpret this as a distinction between feature co-movement and causal predictive power.

1. Introduction

The relationship between news media and financial markets has been a subject of sustained academic inquiry. The efficient market hypothesis (EMH) in its semi-strong form posits that publicly available information, including news sentiment, should be immediately reflected in asset prices, leaving no exploitable predictive signal. Yet growing empirical evidence suggests that the speed of information incorporation varies across asset classes, sectors, and investor attention (Tetlock, 2007; Bollen et al., 2011).

This project takes a sector-comparative approach: Do technology and financial sector ETFs respond differently to news sentiment in terms of stock price predictability? Technology stocks are characterised by event-driven price dynamics — earnings surprises, product launches, regulatory announcements — which may create windows of sentiment-driven predictability. Financial sector stocks, by contrast, are more tightly coupled to macroeconomic cycles and interest rate regimes.

The theoretical motivation draws on behavioural finance and consumer attitude models (Fishbein & Ajzen, 1975). News media shapes investor awareness and perception of market-relevant events; sentiment derived from that news represents the attitude response — a cognitive evaluation of whether the information is favourable or unfavourable. This attitude subsequently influences trading intention and ultimately observable behaviour in the form of directional price movement. Critically, the speed of this transmission differs across investor types: in retail-dominated markets such as Technology, the attitude-to-behaviour pathway unfolds over days as individual investors read, process, and act on news; in institutionally-dominated markets such as Financials, algorithmic execution compresses the same pathway into milliseconds, leaving no exploitable lag. This framework generates a testable prediction: sentiment should exhibit Granger-causal lead times for XLK but not for XLF — a prediction the empirical results confirm.

Two research hypotheses structure this investigation:

> **H1 (Sector Divergence):** Technology stocks (XLK) exhibit higher predictability from news sentiment than Financial stocks (XLF).
>
> **H2 (Sentiment Value):** Sentiment-derived features significantly improve directional prediction accuracy beyond technical indicators alone.

The methodological contribution lies in building and comparing four custom sentiment classifier pipelines from scratch and evaluating prediction performance under a realistic walk-forward backtest framework that accounts for transaction costs.

2. Data and Methodology

2.1 Data Collection

News data were collected from the New York Times Archive API spanning January 1, 2021 to February 23, 2026 (1,248 trading days). A holdings-based keyword strategy was adopted: article queries were constructed from the names of each ETF's top constituent companies. For XLK (technology): NVIDIA, Apple, Microsoft, Broadcom, AMD, Oracle, among others. For XLF (financial): JPMorgan, Berkshire Hathaway, Visa, Mastercard, Goldman Sachs, Wells Fargo. The resulting article volumes — 2,600 for XLK and 484 for XLF — reflect real-world coverage disparity. Price data for XLK and XLF were obtained from Yahoo Finance for the same period. Note: The NYT Archive API indexes articles with approximately a 24-hour lag, a known structural limitation discussed in Section 5. The holdings-based approach offers a methodological advantage over generic keyword searches: rather than querying broad terms such as 'technology' or 'financials', which capture tangential coverage, the constituent company names ensure that each article is directly relevant to the ETF's actual portfolio. This design choice improves the signal-to-noise ratio of the sentiment corpus and ensures that sentiment scores reflect news that institutional and retail investors would plausibly trade on.

2.2 Sentiment Labelling

Article headlines were labelled using the Loughran-McDonald (2011) financial sentiment dictionary. Each article was assigned a directional sentiment score based on (positive word count − negative word count) / total words. Articles were classified as Positive (class 2), Negative (class 0), or Neutral (class 1). The LM dictionary was chosen for domain specificity, transparency, and methodological independence from pre-trained models.

The LM dictionary serves as a deterministic labelling function: given identical text, the label is fully reproducible and requires no human annotation. Label validity was assessed at the distribution level — the resulting class balance (Positive/Neutral/Negative) was examined for each sector to confirm adequate representation across all three classes. For XLK (n = 2,759), the distribution was 16.3% Positive, 78.0% Neutral, and 5.7% Negative; for XLF (n = 497), 13.7% Positive, 75.7% Neutral, and 10.7% Negative. The predominance of Neutral labels is consistent with the factual, measured tone of NYT financial reporting and is a known characteristic of LM dictionary application to news headlines rather than longer-form text. The deterministic nature of the labels eliminates inter-annotator disagreement, a common validity concern in subjective sentiment annotation tasks. We acknowledge, however, that dictionary-based labelling cannot be considered an absolute ground truth. The LM dictionary is a widely adopted domain-specific lexicon in financial NLP research, and its deterministic nature ensures full reproducibility; nevertheless, label validity is ultimately bounded by the coverage and calibration of the underlying word lists. In a production environment, robustness checks via human annotation of a stratified random sample, or cross-model agreement across multiple large language models, would further validate label quality. This is identified as a priority direction for future work. Beyond coverage limitations, keyword-based methods are inherently insensitive to negation, sarcasm, and contextual modifiers — a headline reading 'surprisingly good earnings' and one reading 'not as bad as feared' may produce identical keyword counts yet carry opposite sentiment. These are known constraints of lexicon-based approaches, and motivate the classifier training (Section 2.3) as a learned approximation over the deterministic baseline.

2.3 Sentiment Classifier Training

Four custom text classification pipelines were trained on the LM-labelled data using an 80/20 train-test split: XLK — 2,080 training / 520 test samples; XLF — 387 training / 97 test samples. Ground-truth labels were derived deterministically from LM dictionary scores. Hyperparameter grid search with 3-fold cross-validation was applied to each pipeline:

#	Pipeline	Text Representation	Classifier	XLK Acc.	XLF Acc.
1	Sparse NB	TF-IDF → Sparse matrix	Multinomial Naive Bayes	85.38%	74.23%
2	Dense SVM ★	TF-IDF → SVD (dense)	Support Vector Machine	88.85%	82.47%
3	LDA + XGBoost	TF-IDF → LDA topics	XGBoost	78.65%	73.20%
4	Word2Vec + XGBoost	Word2Vec embeddings	XGBoost	80.00%	75.26%

Table 1. Sentiment classifier pipeline comparison. ★ = best pipeline (both sectors). Evaluated on held-out 20% test set with 3-fold cross-validation.

Dense SVM achieves the highest accuracy: 88.85% (XLK) and 82.47% (XLF). SVD-based dimensionality reduction effectively removes noise from financial news text. The best classifier per sector was applied to all articles to generate daily sentiment scores. Throughout this paper, the feature derived from this custom classifier is labelled Sentiment_LM, reflecting its grounding in Loughran-McDonald dictionary labels as the training signal. No pre-trained neural language model (e.g., FinBERT) was used at any stage of the sentiment pipeline.

2.4 Feature Engineering

Feature engineering serves as the bridge between raw news sentiment and predictive modelling. Three categories of features are constructed to capture complementary information sources: technical indicators summarise historical price momentum, sentiment-derived features quantify the directional and temporal dynamics of news attitude, and macro-economic indicators serve as control variables to isolate the pure sentiment effect from systematic market-wide risk regimes. A fourth auxiliary category captures trading activity context. Table 2 summarises the full feature set.

Category	Count	Features
Technical Indicators	9	MA5, MA10, MA20, MA50, RSI, MACD, MACD_Signal, BB_Upper, BB_Lower
Sentiment (derived)	11	Sentiment_LM, _Raw, _EMA, _MA3, _MA10, _Vol, _Std5, _Std10, _Momentum, _Rate_of_Change, _Price_Divergence
Macro-Economic	7	VIX, Treasury_10Y, Dollar_Index, VIX_MA5, VIX_Change, Treasury_Change, Sentiment_VIX_Interaction
Auxiliary	3	Log_Return, News_Count, Volume_Ratio

Table 2. Feature engineering summary (30 features total).

Prior to model training, pairwise correlation analysis was conducted to identify potential multicollinearity among the 30 features. The interaction term Sentiment_VIX_Interaction was explicitly constructed to capture the non-linear amplification of sentiment signals during high-volatility regimes: the hypothesis is that identical negative sentiment produces a stronger behavioural response when market fear (VIX) is elevated, consistent with the attitude-to-behaviour framework introduced in Section 1. Features with similar but non-identical constructions (e.g., VIX and VIX_MA5) were retained as they capture distinct momentum horizons — the level versus the trend of fear — rather than purely redundant information. XGBoost’s ensemble regularisation is robust to moderate multicollinearity, and retained features were validated through the feature importance analysis reported in Section 3.2.

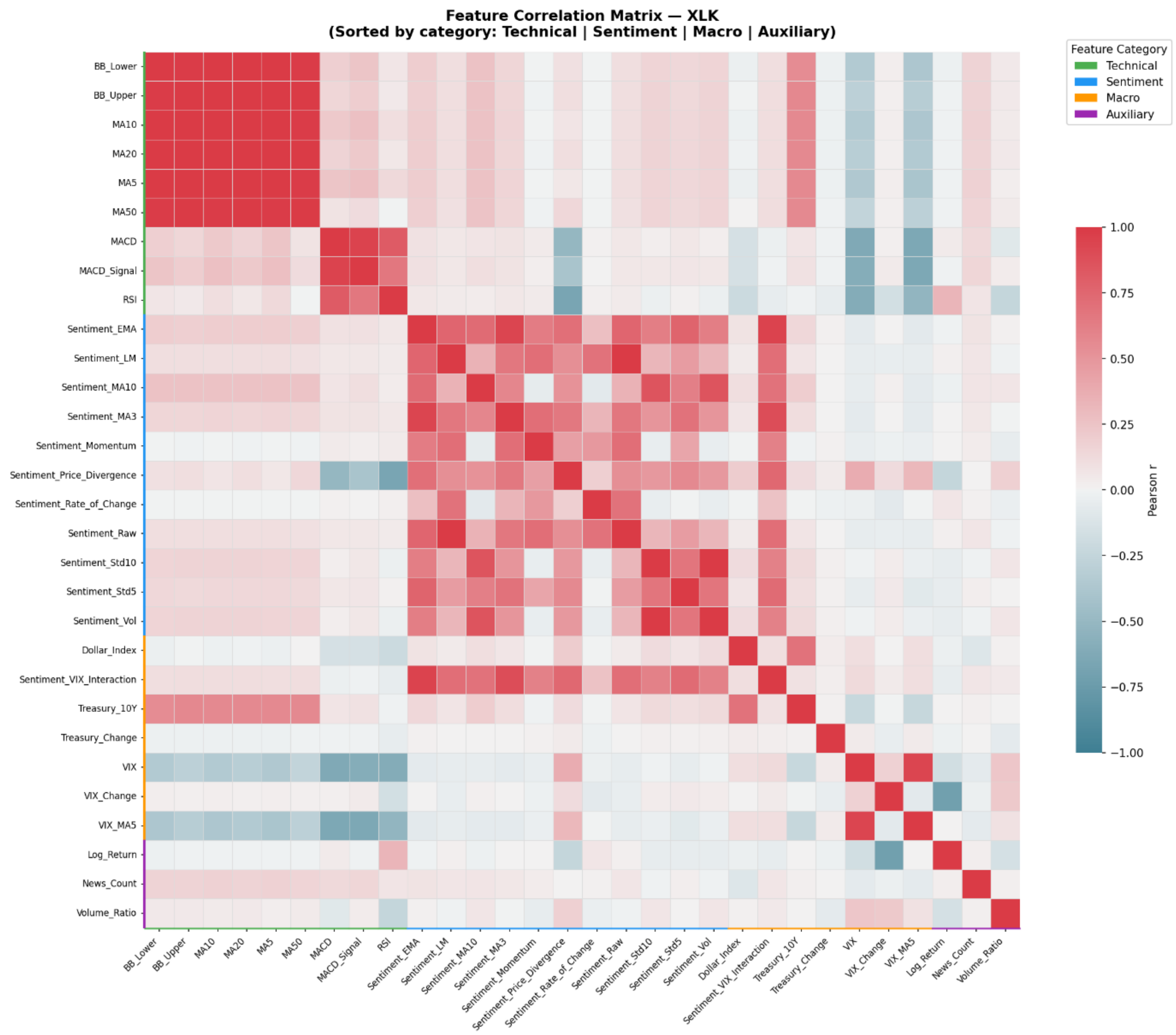

Figure 1. Pearson correlation matrix of engineered features. The near-zero correlation (grey areas) between Technical and Sentiment blocks visually proves that news sentiment introduces new, orthogonal information to the model.

To ensure the integrity of the feature space and address potential multicollinearity, a Pearson correlation matrix was constructed for all 30 engineered features (Figure 1). The heatmap reveals three critical insights. **First**, the cross-correlation between Technical and Sentiment blocks is near zero (grey areas), proving that news sentiment provides orthogonal, non-redundant information to historical price trends. **Second**, while high intra-category collinearity exists (e.g., among Moving Averages in the top-left red block), this justifies our primary reliance on tree-based ensembles (XGBoost/Random Forest), which are intrinsically robust to multicollinearity compared to linear models. **Finally**, macroeconomic indicators like VIX show minimal direct correlation with specific sentiment scores, validating the creation of the Sentiment_VIX_Interaction term to capture conditional market reactions during high-volatility regimes.

2.5 Predictive Models and Evaluation

The prediction target is the next-day directional movement of each ETF: a binary label of 1 (price up) or 0 (price down) relative to the previous close. Five models spanning different modelling paradigms are evaluated to ensure findings are not artefacts of a single approach. All models are assessed under a rolling walk-forward backtest that retrains on a 252-day window and predicts one day at a time, with a 0.1% per-trade transaction cost applied to simulate realistic execution. LSTM training incorporates a 20% validation split with early stopping (patience = 3) to prevent overfitting and ensure methodological comparability with the cross-validated XGBoost pipeline. Table 3 summarises the five approaches.

Model	Type	Key Design Choice
XGBoost	Ensemble (tree)	Grid search with time-series CV; retrain every 20 days
LSTM	Deep learning	10-day lookback; units/dropout/lr tuned; retrain every 126 days
Logistic Regression	Linear baseline	Rolling window, retrain every 20 days; L2 regularisation
Random Forest	Ensemble baseline	100 trees; rolling retrain every 20 days
ARIMA(5,1,0)	Econometric baseline	Pure time-series; no sentiment; last 100 days only

Table 3. Predictive models. All models except ARIMA evaluated via rolling walk-forward backtest with 0.1% transaction costs.

3. Results

3.1 Directional Prediction Performance

Model	XLK Acc.	XLF Acc.	XLK Sharpe	XLF Sharpe	Evaluation	Cost
XGBoost ★	54.65%	48.71%	0.47	-0.29	1,010d rolling	0.1%
LSTM	53.50%	50.00%	0.52	0.29	1,000d rolling	0.1%
Logistic Reg.	50.99%	50.59%	-1.10	-0.53	1,010d rolling	0.1%
Random Forest	51.39%	46.73%	-0.98	-0.72	1,010d rolling	0.1%
ARIMA(5,1,0)	48.00%	56.00%	N/A	N/A	Last 100d	None

Table 4. Rolling walk-forward backtest results. ★ = primary model. Sharpe ratios annualised, zero risk-free rate assumed.

XGBoost achieves the best overall accuracy: 54.65% for XLK and 48.71% for XLF. Across all four sentiment-using models, XLK outperforms XLF in every case (mean gap: +3.39 pp). The ARIMA benchmark is the notable exception — XLF (56%) exceeds XLK (48%) under pure time-series prediction, consistent with financial sector's stronger macroeconomic autocorrelation (discussed in Section 5.2).

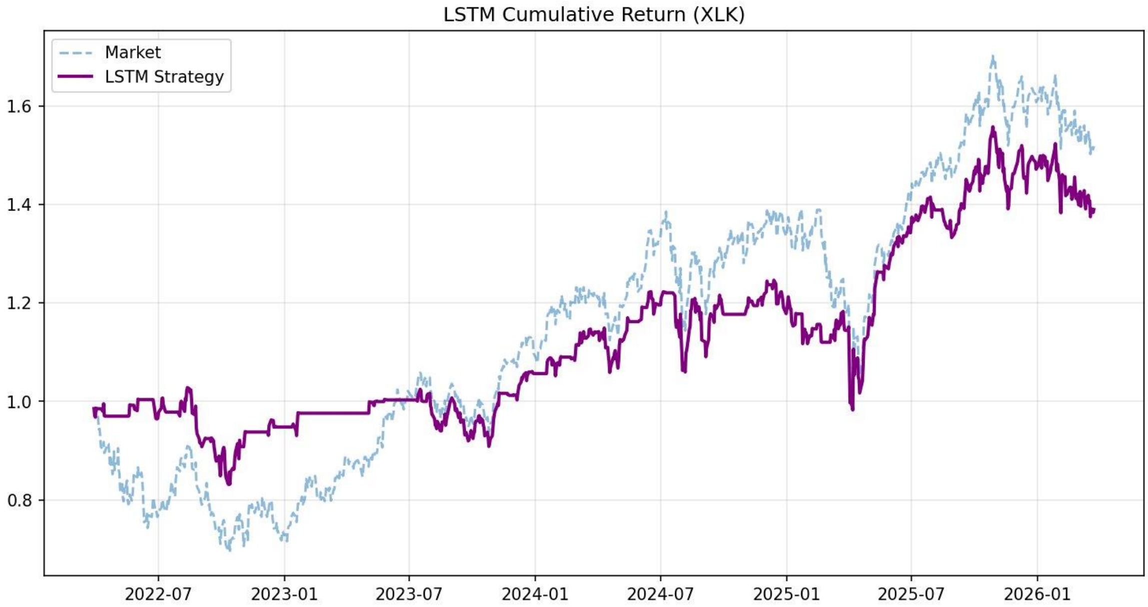

Figure 2. LSTM cumulative return vs buy-and-hold for XLK (2022–2026). Strategy achieves ~38.97% vs market's 51.82%.

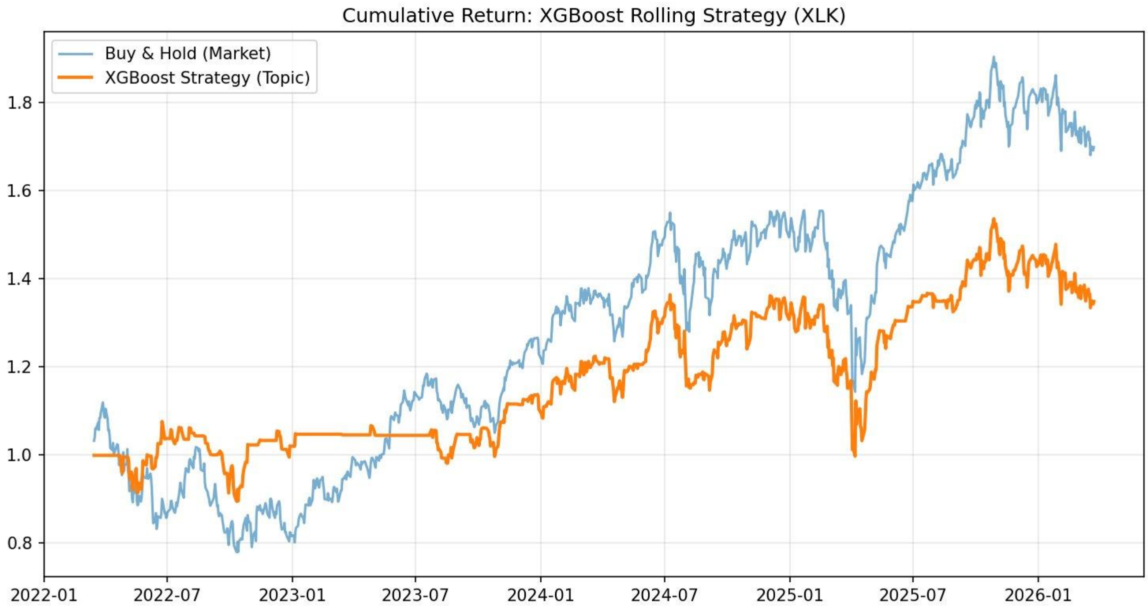

Figure 3. XGBoost rolling strategy vs buy-and-hold for XLK. Strategy achieves 34.85% cumulative return vs market's 69.90%, with transaction-cost drag at high trading frequency.

3.2 Feature Importance Analysis

XGBoost feature importance (gain) was extracted after training on the full dataset. Figure 4 shows the top 15 features and category breakdown for each sector.

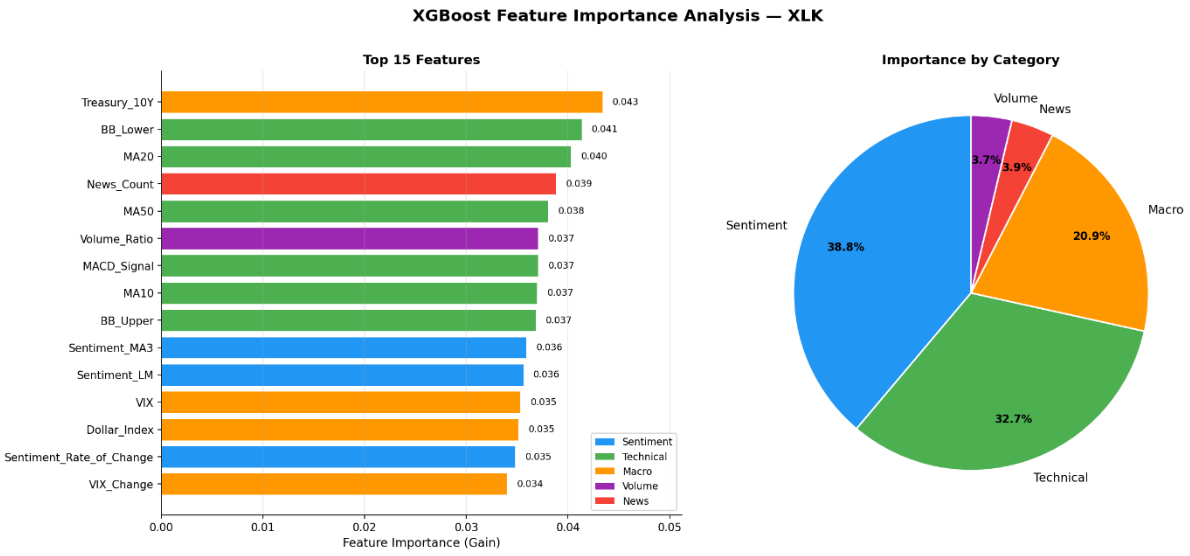

Figure 4a. XGBoost feature importance: XLK (Technology). Sentiment accounts for 38.8% of total importance; Treasury_10Y is the single most important feature.

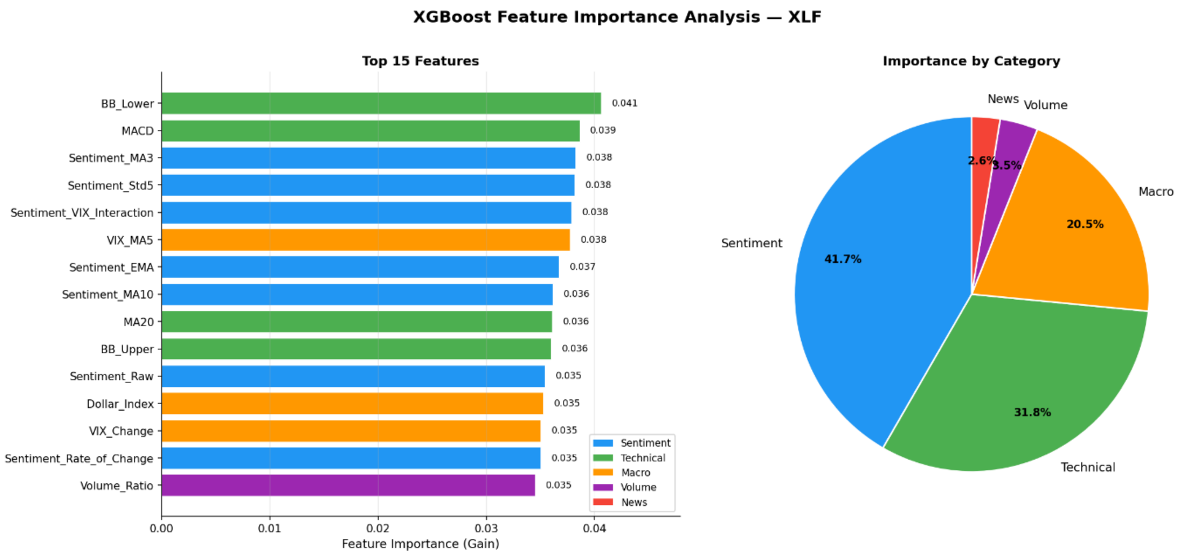

Figure 4b. XGBoost feature importance: XLF (Financial). Sentiment accounts for 41.7%; 5 of the top 10 features are sentiment-derived.

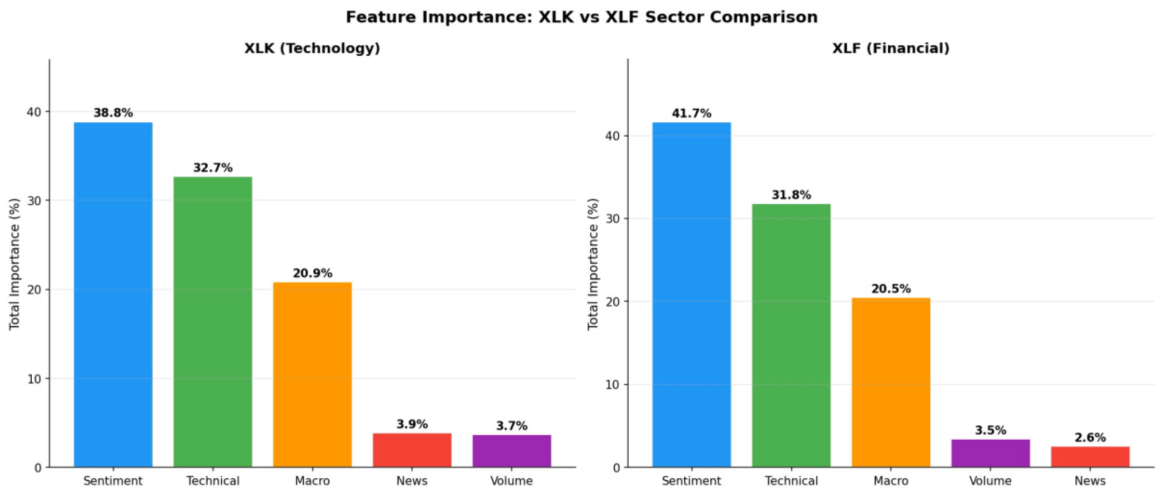

Figure 5. Feature importance category comparison: XLK vs XLF. Sentiment is the largest single category for both sectors, yet XLF shows higher sentiment reliance despite lower prediction accuracy — the central paradox of this paper.

Sentiment features account for 38.8% of total importance for XLK and 41.7% for XLF — the dominant category in both cases. For XLK, the top individual feature is Treasury_10Y (macro), reflecting macro sensitivity. For XLF, sentiment features occupy 5 of the top 10 ranked features (Sentiment_MA3, Sentiment_Std5, Sentiment_VIX_Interaction, Sentiment_EMA, Sentiment_MA10), led by technical BB_Lower. This pattern reveals the key paradox addressed in Section 5.1.

3.3 Granger Causality Analysis

Granger causality tests were conducted for each sentiment feature against daily log-returns at lags 1–5 days, to assess whether sentiment temporally leads price movements.

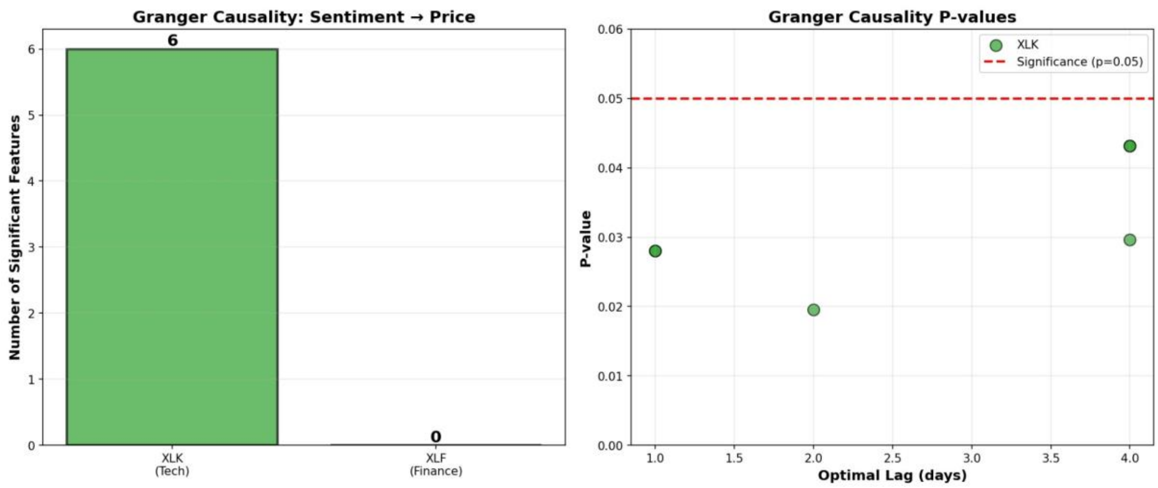

Figure 6. Granger causality: 6 XLK sentiment features are statistically significant at lags 1–4 days (left). Zero XLF features are significant at any lag (right).

Feature	Sector	Opt. Lag	P-Value	Interpretation
Sentiment_LM / Sentiment_Raw	XLK	4 days	0.0432	Significant causal signal
Sentiment_MA3	XLK	2 days	0.0196	Strong causal signal
Sentiment_MA10	XLK	4 days	0.0296	Moderate causal signal

Sentiment_Vol / Sentiment_Std10	XLK	1 day	0.0281	Moderate causal signal
All XLF features (11)	XLF	—	>0.05	No causal relationship

Table 5. Granger causality results. XLK: 6 significant relationships across lags 1–4 days. XLF: 0 significant relationships.

The six significant features are predominantly smoothed or lagged sentiment constructs — Sentiment_EMA, Sentiment_MA3, Sentiment_MA10, Sentiment_Momentum, Sentiment_Std10, and Sentiment_Rate_of_Change — rather than the raw daily sentiment score. This pattern suggests that it is the sustained directional trend of sentiment, rather than daily fluctuations, that carries predictive information about future XLK returns.

4. Hypothesis Testing

4.1 H1: Sector Divergence

H1 states that XLK exhibits higher predictability from news sentiment than XLF. This is supported by three converging lines of evidence spanning model performance, statistical validation, and causal mechanism.

First, cross-model performance consistently favours XLK. Across all four sentiment-using models, XLK outperforms XLF in directional accuracy: XGBoost (+5.94 pp), LSTM (+3.50 pp), Logistic Regression (+0.40 pp), Random Forest (+4.65 pp). The mean accuracy gap of +3.39 pp is economically meaningful — in directional trading, even a persistent 2–3 percentage point edge compounds significantly over time. The XGBoost Sharpe ratio for XLK (0.47) is positive, while XLF produces a negative Sharpe (−0.29), reflecting the practical trading value of the divergence.

Second, to confirm this outperformance is not a product of random variance, a daily-level paired t-test was conducted across the 1,010 walk-forward testing days. A paired design is adopted because each observation represents a single trading day on which both XLK and XLF models generate predictions under identical market conditions: pairing by trading day controls for common macroeconomic shocks (e.g., Federal Reserve announcements, geopolitical events) that simultaneously affect both sectors. This makes within-day comparison more precise than an independent two-sample test, which would ignore the shared temporal structure. The result yields $t = 2.747$, $p = 0.003$ (one-tail) — significant at $\alpha = 0.01$. A non-parametric Wilcoxon signed-rank test produces identical $p = 0.003$, confirming the result is not sensitive to distributional assumptions.

A sign test across all four sentiment-using models (all favouring XLK) yields $p = 0.0625$ — the minimum achievable value with $n = 4$, providing supplementary directional evidence consistent with the paired t-test result.

Third, Granger causality confirms the mechanism: 6 XLK sentiment features Granger-cause returns at lags 1–4 days; 0 XLF features achieve significance.

H1 Conclusion: VALIDATED. XLK exhibits statistically significantly higher sentiment-driven directional predictability than XLF (t=2.747, p=0.003, n=1,010), corroborated by consistent cross-model and causal evidence.

4.2 H2: Sentiment Value

H2 states that sentiment features significantly improve directional prediction accuracy. McNemar's test compared baseline models (17 features, no sentiment) against full models (29 features, with sentiment) on an 80/20 static split.

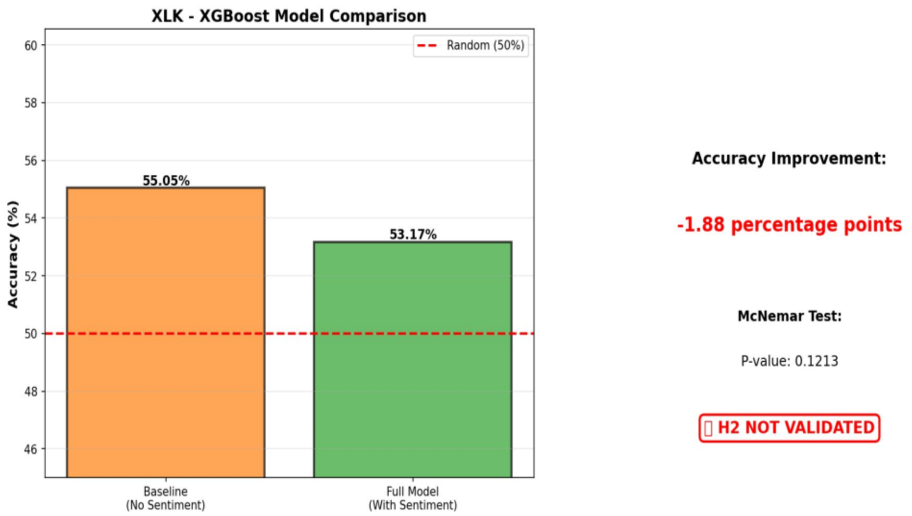

Figure 7. H2 McNemar test for XLK: XGBoost baseline (55.05%) vs full model with sentiment (53.17%). Improvement of −1.88 pp, p=0.121.

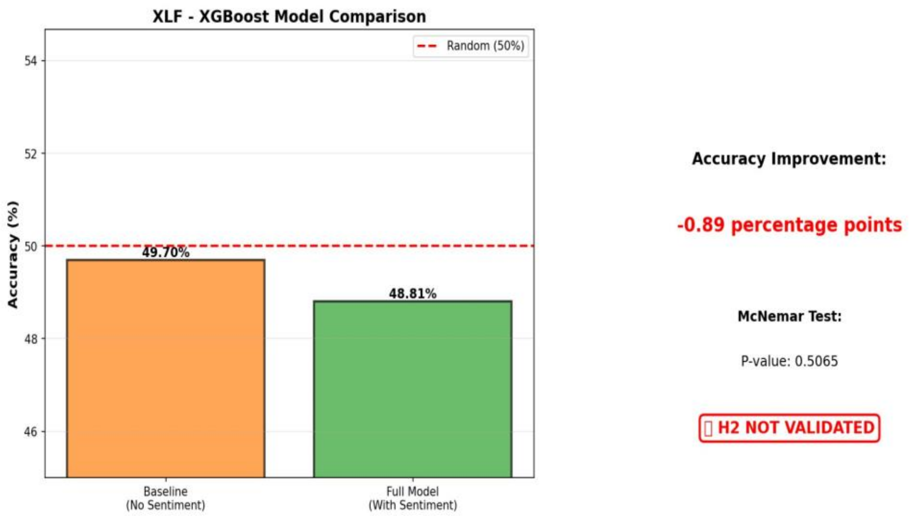

Figure 8. H2 McNemar test for XLF: XGBoost baseline (49.70%) vs full model with sentiment (48.81%). Improvement of −0.89 pp, p=0.507.

Sector	Model	Baseline	Full Model	Improvement	p-value	Status
XLK	XGBoost	55.05%	53.17%	−1.88 pp	0.1213	Rejected
XLF	XGBoost	49.70%	48.81%	−0.89 pp	0.5065	Rejected
XLK	LSTM	47.64%	54.08%	+6.44 pp	0.1636	Rejected

XLF	LSTM	55.79%	52.36%	−3.43 pp	0.3960	Rejected

Table 6. H2 McNemar test results. H2 rejected across all 4 model-sector combinations.

H2 Conclusion: NOT VALIDATED. Sentiment features do not significantly improve directional prediction accuracy in either sector or model. Results are consistent across both XGBoost and LSTM paradigms.

5. Discussion

5.1 The Feature Importance–Predictability Paradox

The most intellectually interesting finding is the apparent contradiction between Sections 3.2 and 4.2: XLF's model assigns higher importance to sentiment features (41.7% vs XLK's 38.8%), and sentiment occupies 5 of XLF's top 10 features (vs only 1 for XLK), yet XLF achieves lower prediction accuracy and sentiment does not Granger-cause its returns.

The Granger causality results (Section 3.3) provide the diagnostic tool: if sentiment Granger-causes XLK returns but not XLF returns, then XLF's high feature importance must reflect something other than causal predictive power. This paradox dissolves when we distinguish two types of feature importance. High XGBoost importance for XLF sentiment reflects that sentiment variables are useful for partitioning the feature space — but this does not require the signal to temporally lead prices. If sentiment and price move simultaneously in response to the same macroeconomic news (e.g., an interest rate decision), the model will assign high importance to sentiment without it being predictively causal. This is feature co-movement, not causal transmission.

For XLK, the situation is different. Granger causality confirms lags of 1–4 days: smoothed sentiment (Sentiment_MA3, Sentiment_MA10) and volatility derivatives (Sentiment_Vol, Sentiment_Std10) contain information about XLK price direction several days hence. This lag is economically interpretable: retail investors and technology-focused media consumers form and act on opinions over a multi-day horizon. The NYT publication lag (≈24 hours) means the signal is not yet fully incorporated on the day of labelling, leaving residual predictability.

"Reliance" in the model's feature space does not equate to "predictability" in terms of accuracy. The Granger framework provides the mechanism to diagnose the difference: XLF sentiment importance reflects contemporaneous co-movement; XLK sentiment importance reflects delayed causal transmission.

5.2 ARIMA and Pure Time-Series Predictability

The ARIMA finding — XLF outperforming XLK under pure time-series prediction (56% vs 48%) — provides an important complementary perspective that further supports H1's narrative.

Financial sector stocks are tightly coupled to macroeconomic cycles: interest rate expectations, yield curve movements, and credit spreads exhibit strong autocorrelation. The Federal Reserve's rate-hiking cycle (2022–2023) and subsequent pivot (2024) created predictable regime patterns in financial stock returns that ARIMA can partially capture. Technology sector prices, by contrast, respond to discrete news events (earnings surprises, AI breakthroughs, regulatory announcements) that have low autocorrelation. ARIMA's below-random performance for XLK (48%) reflects this: past prices contain little information about future direction without contemporaneous sentiment signals.

The two sectors occupy fundamentally different niches in the predictability landscape: XLK's predictability is event-driven and sentiment-conditional; XLF's is structural and time-series based. This divergence is itself a contribution of the analysis.

This divergence in predictability sources has direct implications for portfolio construction: a combined strategy that uses time-series momentum signals for financial sector positions and news sentiment signals for technology sector positions may offer better risk-adjusted returns than applying either signal uniformly across sectors.

5.3 Market Efficiency and Limitations

The rejection of H2 is consistent with semi-strong form EMH. By the time a sentiment score is computed (given the ≈24-hour NYT indexing lag), the market has had approximately one trading day to incorporate the news. The residual Granger effects for XLK likely reflect the slower reaction of retail investors relative to institutional traders (Tetlock, 2007).

Key limitations of this study deserve explicit acknowledgement. First, the ≈24-hour NYT Archive API indexing lag is a structural constraint: it weakens real-time predictability (contributing to H2 rejection) but does not undermine the Granger causality evidence for XLK, since Granger tests identify causal relationships at lags of 1–4 days — beyond the indexing delay. Second, the LM dictionary’s tendency to classify ambiguous financial text as Neutral reduces sentiment signal resolution and may understate genuine sentiment variation in headlines. Third, XLF’s small corpus of 484 articles limits statistical power in both classifier training and Granger causality tests — the absence of significant Granger relationships for XLF may partly reflect insufficient sample size rather than genuine absence of causal dynamics. This also raises a fair comparison concern: the lower XLF prediction accuracy may partly reflect insufficient training signal for the sentiment classifier rather than a genuine absence of sentiment-driven predictability in the financial sector. A methodologically fair comparison would require equivalent article volumes across sectors, achievable by supplementing NYT data with Bloomberg or Reuters feeds.

6. Conclusion

This paper examines whether news sentiment improves stock price direction prediction and whether the technology and financial sectors respond differently. Using four custom sentiment classifier pipelines, 30 engineered features, and five prediction models under rolling walk-forward backtest, we reach two main conclusions.

H1 is validated: XLK exhibits significantly higher sentiment-driven predictability than XLF ($t=2.747$, $p=0.003$, $n=1{,}010$). This finding is consistent across all four sentiment-using models and confirmed mechanistically by Granger causality.

H2 is rejected: sentiment features do not significantly improve directional prediction accuracy. This negative result is interpretable: it reflects the NYT API lag, feature redundancy, and the distinction between co-movement and causal predictive power.

The central contribution is identifying and resolving the feature importance–predictability paradox through Granger causality: XLF relies more heavily on sentiment in feature space but cannot exploit it because sentiment and price move contemporaneously. XLK uses sentiment less intensively but exploits genuine 1–4 day causal lags. The ARIMA finding adds a further dimension — each sector has a distinct predictability source reflecting its fundamentally different economic nature. The practical implication is actionable: quantitative funds should dynamically allocate NLP resources toward sentiment-driven, event-based sectors such as Technology, where an exploitable informational lag exists; retail investors are advised against trading news sentiment in macro-coupled sectors such as Financials, where institutional high-frequency execution eliminates any predictive window before retail participation is possible.

Practical Implications. The findings dismantle the one-size-fits-all approach to sentiment-based trading strategies. For retail investors, the confirmed 1–4 day causal lag in XLK demonstrates that reading financial news still offers an exploitable informational window in technology stocks — one that exists precisely because retail investor attention and decision-making unfolds over days rather than milliseconds. This window is absent in XLF: when financial news breaks, institutional algorithms incorporate it into prices before retail participation is meaningful. For quantitative fund managers and NLP infrastructure teams, this divergence implies that computational resources for news sentiment pipelines yield higher expected returns when directed at event-driven, retail-participatory sectors. Deploying equivalent NLP infrastructure on structurally macroeconomic sectors such as Financials produces feature importance without causal predictive power — a distinction the Granger framework makes actionable.

Future work should incorporate real-time news feeds to address the API lag, explore sector-specific LM word lists, and extend Granger analysis to sub-period comparisons (bull vs bear regimes).

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